Project: Worked examples
Title: Motor and actuator sizing
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Torque, speed and power a motor needs to move a load through a mechanism on a point-to-point cycle: the load reflected to the motor shaft, the acceleration from the motion profile, peak torque during acceleration, RMS torque over the cycle (thermal duty), peak speed, inertia ratio, and a check of a candidate motor's rated and peak torque and speed with a safety margin. A rotary load that runs continuously (a pump, a fan) is sized instead from its steady torque and speed and a run-up time.
Linear mechanisms: k = metres per motor radian (screw p/(2π i); belt D/(2 i); rack D_p/(2 i)); J_refl = m k²/η; T_load = F k/η with F = m g (sin θ + μ cos θ) + F_p. Rotary: J_refl = J_L/(i² η), T_load = T_L/(i η). Trapezoidal 1/3–1/3–1/3 (adjustable): v_pk = s/(t_move − t_acc), a = v_pk/t_acc; S-curve: sine acceleration with a_pk = (π/2) a_trap and the same v_pk. T_acc = J_total α; T_pk = T_acc + T_load; T_rms = √(Σ T_i² t_i / t_cycle) including the dwell. RMS mechanical power (the thermal-equivalent continuous power a supply sees; regeneration counts as power): P_rms = ω_pk √{[t_acc·(2/3)(T_load² + T_acc²) + T_load² t_const]/t_cycle} (trapezoid), with t_acc·[(3/4) T_load² + (5/16) T_acc²] for the S-curve; the instantaneous peak P_pk is not a continuous demand. Continuous rotary duty: ω = 2π n_L i/60; T_load = T_L/(i η); T_rms = T_load; T_pk = J_total ω/t_start + T_load (full load during the run-up, conservative); n_pk = n_L i; P_cont = T_load ω; P_rms = P_cont.
Ref: Hughes & Drury, Electric Motors and Drives Ch. on motor and drive selection — Load torque, inertia referred to the motor, duty cycles
Ref: Manufacturer servo-sizing guides (Kollmorgen, Parker) — Reflected inertia, RMS torque, inertia ratio practice

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mechanism	Mechanism	gearbox rotary		assumption: Sizing convention.	any
i	Gear ratio i (motor : output, 1 = none)	10		spec: Gearbox datasheet PG-10: 10:1, 85 %	1 … 1000
eta	Mechanism efficiency (overall)	0.85		spec: Gearbox datasheet PG-10: 10:1, 85 %	0.3 … 1
J_mech	Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)	0	kg*m^2	assumption: Sizing convention.	0 … 10 kg*m^2
mu	Friction coefficient (guide / bearing)	0.1		default	0 … 1
theta	Incline from horizontal (+ moving uphill)	0	deg	default	-90 … 90 deg
F_p	Process force opposing the move (linear)	0	N	default	0 … 1000000 N

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
J_L	Load inertia at the output (rotary mechanisms)	0.5	kg*m^2	spec: S1-style inertia calc of the 600 mm table with fixtures: 0.5 kg·m²	0 … 10000 kg*m^2
T_L	Constant load torque at the output (rotary)	0	N*m	assumption: Sizing convention.	0 … 100000 N*m
duty_mode	Duty (point-to-point move cycle, or continuous rotary running with a run-up)	point-to-point		default	any
profile	Motion profile	trapezoidal		assumption: Sizing convention.	any
s	Move distance (linear: m; rotary: output revolutions)	0.25		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.0001 … 10000
t_move	Move time	0.3	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.01 … 3600 s
f_acc	Acceleration (and deceleration) fraction of the move time	0.333333		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.05 … 0.5
t_dwell	Dwell between moves	0.7	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0 … 3600 s
J_motor	Candidate motor rotor inertia	0.0005	kg*m^2	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	1e-7 … 10 kg*m^2
T_rated	Candidate motor continuous (rated) torque	3	N*m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N*m
T_peak	Candidate motor peak torque	9	N*m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N*m
n_max	Candidate motor maximum speed	3000	rpm	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	10 … 100000 rpm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
SF	Safety margin on torque	1.25		assumption: Sizing convention.	1 … 3
J_ratio_max	Inertia ratio limit (load : motor), warning	10		assumption: Sizing convention.	1 … 100

3. Assumptions

A1. Rigid transmission; the mechanism efficiency applies to the driving direction and is also used in the deceleration phase (conservative for peak, slightly pessimistic for RMS). (Servo-sizing practice.)
A2. The friction force is Coulomb friction on the moved mass; viscous and breakaway friction are not modelled unless included in F_p. (Load model.)
A3. Continuous duty: the load torque is constant at the stated speed and the motor accelerates the whole inertia linearly to speed in t_start while carrying the full load torque (conservative); the move distance, move time, acceleration fraction, dwell and profile are ignored. (Run-up idealisation for pumps and fans.)
A4. The S-curve profile has sine-shaped acceleration over the same acceleration time, so its peak acceleration is π/2 times the trapezoidal value. (Jerk-limited profile idealisation.)
A5. mechanism (Mechanism) is taken as gearbox rotary. (Sizing convention.)
A6. J_mech (Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)) is taken as 0 kg*m^2. (Sizing convention.)
A7. T_L (Constant load torque at the output (rotary)) is taken as 0 N*m. (Sizing convention.)
A8. profile (Motion profile) is taken as trapezoidal. (Sizing convention.)
A9. SF (Safety margin on torque) is taken as 1.25. (Sizing convention.)
A10. J_ratio_max (Inertia ratio limit (load : motor), warning) is taken as 10. (Sizing convention.)

4. Working

[21] k = 1 / i
= 1 / 10
= 0.1
Output radians per motor radian
[22] T_load = T_L / (i * eta)
= 0 N·m / (10 × 0.85)
= 0 N·m
Load torque at the motor shaft
[23] J_refl = J_L / (i^2 * eta)
= 0.5 kg·m² / (10^2 × 0.85)
= 0.0058824 kg·m²
Load inertia reflected to the motor shaft
[24] theta_m = s * 2 * pi * i
= 0.25 × 2 × π × 10
= 15.708 rad
Motor angle for the move
[25] omega_pk = theta_m / (t_move * (1 - f_acc))
= 15.708 rad / (0.3 s × (1 - 0.33333))
= 78.54 rad/s
Peak motor speed (trapezoid)
[26] a_trap = omega_pk / (f_acc * t_move)
= 78.54 rad/s / (0.33333 × 0.3 s)
= 785.4 rad/s²
Trapezoidal angular acceleration at the motor
[28] a_pk = a_trap
= 785.4 rad/s²
= 785.4 rad/s²

Peak acceleration

```
[29] alpha_pk = a_pk
= 785.4 rad/s²
= 785.4 rad/s²
```

Peak motor angular acceleration

```
[30] J_total = J_motor + J_mech + J_refl
= 0.0005 kg·m² + 0 kg·m² + 0.0058824 kg·m²
= 0.0063824 kg·m²
```

Total inertia at the motor shaft

```
[31] T_acc = J_total * alpha_pk
= 0.0063824 kg·m² × 785.4 rad/s²
= 5.0127 N·m
```

Acceleration torque

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[32] T_pk = T_acc + T_load
= 5.0127 N·m + 0 N·m
= 5.0127 N·m
```

Peak torque (accelerating against the load)

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[33] T_dec = -T_acc + T_load
= -5.0127 N·m + 0 N·m
= -5.0127 N·m
```

Torque during deceleration

```
[34] n_pk = omega_pk
= 78.54 rad/s
= 750 rpm
```

Peak motor speed

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[35] P_pk = T_pk * omega_pk
= 5.0127 N·m × 78.54 rad/s
= 0.3937 kW
```

Peak mechanical power

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[36] t_acc = f_acc * t_move
= 0.33333 × 0.3 s
= 0.1 s
```

Acceleration time

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[37] t_const = t_move - 2 * t_acc
= 0.3 s - 2 × 0.1 s
= 0.1 s
```

Constant-velocity time

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[38] t_cycle = t_move + t_dwell
= 0.3 s + 0.7 s
= 1 s
```

Cycle time

```
[39] P_rms = omega_pk * sqrt((t_acc * (2 / 3) * (T_load^2 + T_acc^2) + T_load^2 * t_const) /
= 78.54 rad/s × sqrt((0.1 s × 2 / 3 × ((0 N·m)^2 + (5.0127 N·m)^2) + (0 N·m)^2 × 0.1 s) / 1 s
= 0.10165 kW
```

RMS mechanical power over the cycle (thermal-equivalent continuous power)

```
[40] T_rms = sqrt((T_pk^2 * t_acc + T_load^2 * t_const + T_dec^2 * t_acc) / t_cycle)
= sqrt(((5.0127 N·m)^2 × 0.1 s + (0 N·m)^2 × 0.1 s + (-5.0127 N·m)^2 × 0.1 s) / 1 s)
= 2.2417 N·m
```

RMS torque over the cycle (thermal)

```
[41] J_ratio = (J_refl + J_mech) / J_motor
= (0.0058824 kg·m² + 0 kg·m²) / 0.0005 kg·m²
= 11.765
```

Inertia ratio (load : motor)

[42] u_peak = SF * T_pk / T_peak
= 1.25 × 5.0127 N·m / 9 N·m
= 0.69621
Peak torque utilisation with safety margin
[43] u_rms = SF * T_rms / T_rated
= 1.25 × 2.2417 N·m / 3 N·m
= 0.93406
Continuous torque utilisation with safety margin
[44] u_speed = n_pk / n_max
= 750 rpm / 3000 rpm
= 0.25
Speed utilisation
[45] duty = T_rms / T_rated
= 2.2417 N·m / 3 N·m
= 0.74725
Thermal duty (without margin)

5. Results

SYMBOL	VALUE	DESCRIPTION
k	0.1	Transmission constant
T_load	0 N·m	Load torque at motor
J_refl	0.0058824 kg·m²	Reflected load inertia
J_total	0.0063824 kg·m²	Total inertia
theta_m	15.708 rad	Motor angle
a_pk	785.4 rad/s²	Peak acceleration
alpha_pk	785.4 rad/s²	Motor angular acceleration
T_acc	5.0127 N·m	Acceleration torque
T_pk	5.0127 N·m	Peak torque
T_dec	-5.0127 N·m	Deceleration torque
T_rms	2.2417 N·m	RMS torque
n_pk	750 rpm	Peak speed
P_pk	0.3937 kW	Peak power
P_rms	0.10165 kW	RMS power (continuous equivalent)
J_ratio	11.765	Inertia ratio
duty	0.74725	Thermal duty
u_peak	0.69621	Peak utilisation
u_rms	0.93406	Continuous utilisation
u_speed	0.25	Speed utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	RMS torque SF·T_rms ≤ T_rated	u_rms = 0.93406	u_rms ≤ 1	0.934	

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Peak torque SF·T_pk ≤ T_peak	u_peak = 0.69621	u_peak ≤ 1	0.696	
PASS	Speed n_pk ≤ n_max	u_speed = 0.25	u_speed ≤ 1	0.25	

7. Applicability

WARNING — input.mu: mu (Friction coefficient (guide / bearing)) was not entered; the template default 0.1 is used.
WARNING — input.theta: theta (Incline from horizontal (+ moving uphill)) was not entered; the template default 0 deg is used.
WARNING — input.F_p: F_p (Process force opposing the move (linear)) was not entered; the template default 0 N is used.
WARNING — input.duty_mode: duty_mode (Duty (point-to-point move cycle, or continuous rotary running with a run-up)) was not entered; the template default 'point-to-point' is used.
WARNING — inertia-ratio: J_motor: the inertia ratio 11.8 exceeds 10.
WARNING — overhauling: T_dec: the motor brakes during deceleration (negative torque); the drive must absorb the regenerated energy.

8. Verification

Template elec.motor-sizing@1.2.0 Content hash 07f83040d397a49f Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0